

Smartphone Market Analysis

Business Requirement

Business Scenario

A leading mobile retail company wants to analyze the smartphone market to understand customer preferences and identify which smartphone brands provide the best value for money. With hundreds of smartphone models available from different manufacturers, it has become difficult for customers to compare devices based on price, RAM, processor, battery capacity, camera specifications, and screen size.

As a **Junior Data Analyst**, your responsibility is to analyze the smartphone dataset using Python and Pandas. You will perform data cleaning, exploratory data analysis (EDA), and visualization to identify market trends, compare smartphone brands, and recommend the best smartphones based on their specifications and pricing.

Using Python and Pandas, perform data cleaning, exploratory data analysis (EDA), and visualization to answer key business questions and provide meaningful recommendations.

Project Objective

Analyze the smartphone dataset to:

- Understand the smartphone market across different brands.
 - Compare smartphone specifications such as RAM, processor, battery, and cameras.
 - Analyze pricing trends in the Indian market.
 - Identify smartphones that offer the best value for money.
 - Generate business insights and recommendations.
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Dataset Description

The dataset contains information about **930 smartphone models** launched by different mobile brands across multiple countries. For this project, students should primarily focus on the **Indian market** using the **Launched Price (India)** column.

Dataset Features

Column	Description
Company Name	Smartphone Brand
Model Name	Smartphone Model
Mobile Weight	Weight of the Phone
RAM	RAM Capacity
Front Camera	Front Camera Specification
Back Camera	Rear Camera Specification
Processor	Mobile Processor
Battery Capacity	Battery Capacity
Screen Size	Display Size
Launched Price (Pakistan)	Launch Price in Pakistan
Launched Price (India)	Launch Price in India
Launched Price (China)	Launch Price in China
Launched Price (USA)	Launch Price in USA
Launched Price (Dubai)	Launch Price in Dubai
Launched Year	Launch Year

Tasks to Perform

Task 1 – Data Loading

- Import the required libraries.
- Load the dataset using Pandas.
- Display the first five records.
- Display the last five records.
- Find the number of rows and columns.
- Display all column names.

- Check the data types of each column.
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Task 2 – Data Exploration

Perform the following analysis:

- Check for missing values.
 - Check duplicate records.
 - Display summary statistics.
 - Find the number of unique smartphone brands.
 - Find the number of unique processors.
 - Find the number of unique launch years.
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Task 3 – Data Cleaning

Perform the following operations:

- Remove duplicate records (if any).
 - Handle missing values.
 - Clean the **Launched Price (India)** column and convert it into numeric format.
 - Clean the **Battery Capacity**, **RAM**, and **Screen Size** columns (if required).
 - Ensure the **Launched Year** column has the correct data type.
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Business Questions

Answer the following questions using Python and Pandas.

Brand Analysis

- Which company has launched the highest number of smartphones?
 - Display the Top 10 smartphone brands based on the number of models.
 - Which brand has the highest average launch price in India?
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Price Analysis

- What is the average smartphone launch price in India?
- Which are the Top 10 most expensive smartphones launched in India?

- Which are the Top 10 least expensive smartphones launched in India?
 - Which smartphones were launched under ₹30,000?
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Hardware Analysis

- Which smartphones have the highest RAM?
 - Which smartphones have the largest battery capacity?
 - Which processors are used most frequently?
 - Which processor appears in the highest-priced smartphones?
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Camera Analysis

- Which smartphones have the highest front camera specifications?
 - Which smartphones have the highest back camera specifications?
 - Which brands offer the best camera specifications?
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Display Analysis

- What is the average screen size of smartphones?
 - Which smartphones have the largest displays?
 - Which brands offer larger screen sizes on average?
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Launch Year Analysis

- How many smartphones were launched each year?
 - Which year saw the highest number of smartphone launches?
 - Has the average launch price increased over the years?
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Data Visualization

Create **at least five charts** using Matplotlib.

Mandatory Charts

- Top 10 Smartphone Brands by Number of Models (Bar Chart)
- Average Launch Price by Brand (Bar Chart)
- Smartphone Launches by Year (Line Chart)
- Launch Price Distribution in India (Histogram)
- Top 10 Smartphones by Battery Capacity (Horizontal Bar Chart)

Optional Chart

- Processor Distribution (Pie Chart)
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Business Insights

Write **at least 5 business insights** based on your analysis.

Examples:

- Which smartphone brands dominate the market?
 - Which brands offer the most affordable smartphones?
 - Which processor is most commonly used?
 - Which brands provide larger batteries and higher RAM?
 - Which launch year witnessed the highest number of smartphone releases?
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Business Recommendations

Provide **at least 3 recommendations** for the mobile retail company.

Example recommendations:

- Recommend smartphones that offer the best balance of price and specifications.
 - Focus marketing efforts on brands with competitive pricing and strong hardware features.
 - Promote smartphones under ₹30,000 that provide higher RAM and larger battery capacity.
 - Maintain a diverse inventory by stocking popular brands and processor variants.
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Submission Requirements

Students must submit:

- Python Notebook (.ipynb)
 - Cleaned Dataset (if modified)
 - Report
 - 3 Business Recommendations
 - 5-minute PowerPoint Presentation
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Evaluation Criteria

Criteria	Marks
Data Loading & Exploration	15
Data Cleaning	15
Data Analysis	30
Visualizations	20
Business Insights	10
Recommendations	10
Total	100

Note: Students should primarily analyze the **Indian smartphone market** by using the **Launched Price (India)** column. The remaining country-specific price columns may be used for optional comparative analysis if desired.